Does "Excitation-Inhibition Imbalance" Explain Ketamine's Antidepressant Effect in Major Depressive Disorder?

Summary

Ketamine's clinical antidepressant benefit in Major Depressive Disorder (MDD) and treatment-resistant depression (TRD) is firmly established, but the molecular/cellular mechanism that *produces* that benefit is NOT. The "Excitation-Inhibition (E/I) Imbalance" mechanism asserted in the knowledge base — NMDA receptor (NMDAR) antagonism on GABAergic interneurons causing disinhibition, a glutamate surge, AMPA-receptor activation, and downstream BDNF–mTORC1 synaptogenesis — is the dominant textbook hypothesis. It has partial, largely indirect support, but direct tests in patients are null and multiple strands of evidence show it is, at best, *necessary but not sufficient*, and possibly not causal at all.

The strongest challenge to the E/I account is a convergence of dissociating evidence: (1) other agents that engage NMDARs with similar target engagement — memantine and lanicemine (AZD6765) — repeatedly fail to produce antidepressant benefit in RCTs; (2) the ketamine metabolite (2R,6R)-hydroxynorketamine reproduces antidepressant-like actions in mice *independent of NMDAR inhibition*; (3) a 2026 meta-analysis of patient ^{1}H-MRS studies found no reliable ketamine-induced change in GABA, glutamate, or Glx — the very readouts that define "E/I"; (4) the acute dissociative response (the most direct behavioral NMDAR/E-I signature) does not track clinical response; and (5) human pharmacological dissection with naltrexone shows opioid-receptor blockade can abolish ketamine's antidepressant effect while *sparing* dissociation, implicating a pathway outside the E/I framework. Partial support exists — frontoparietal gamma power (an E/I-related cortical readout) is a promising response biomarker — but it is correlative.

Assessment: the mechanism is best classified as "several competing candidates," with E/I imbalance a leading hypothesis that lacks causal confirmation in patients. Multiple independent 2024–2026 reviews explicitly state the mechanism "remains unclear." The

knowledge base should record the E/I-imbalance mechanism as **one of several candidate mechanisms / an open knowledge gap**, not as established. This report gives the refuting evidence the same prominence as the supporting evidence, as instructed.

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1. Clinical Effect (Context)

Ketamine's clinical benefit in MDD/TRD is well established and is not the subject of dispute here.

A systematic review and meta-analysis of 8 RCTs (n = 414) of **oral ketamine** in depression found a pooled standardized mean difference (SMD) of **−0.62 (95% CI −0.39 to −0.85, Z = 5.24, p < 0.00001)** with moderate certainty, and a remission risk ratio of 2.38 (95% CI 1.19–4.77, p = 0.01).

> *"The depression scores showed a significant decrease after oral ketamine treatment versus placebo, with a SMD of − 0.62 [95% CI= -0.39, -0.85, Z = 5.24, p < 0.00001] with moderate certainty of the evidence."* — PMID: 42155204

Regulatory approval confirms the established efficacy of the esketamine enantiomer:

> *"Esketamine nasal spray was the first glutamate-modulating antidepressant to enter clinical practice and was recently approved as the first and only monotherapy for adults with TRD."* — PMID: 41923438

This is context. The research question concerns the **mechanism**, addressed below.

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2. Candidate Mechanisms

Several distinct molecular/cellular mechanisms have been proposed for ketamine's antidepressant benefit in MDD. The table summarizes each candidate, whether ketamine's action on that target is itself established, and whether the *therapeutic benefit* has been shown to depend on it.

Candidate mechanism	Drug action on target established?	Benefit shown to *depend* on it?	Key evidence
E/I imbalance (NMDAR antagonism on interneurons → disinhibition → glutamate/ AMPA surge)	Yes — ketamine is a confirmed NMDAR antagonist	**No** — direct patient MRS readouts null; NMDAR-engaging analogues fail	F002, F003, F009
BDNF–mTORC1 / AMPA synaptogenesis (downstream node)	Yes (preclinical)	Partially in animals (BDNF-dependent); **not** in patients (BDNF does not track response)	F007, F012
NMDAR-independent HNK metabolite / AMPA activation	Yes (preclinical)	Supported in mice, *independent of NMDAR inhibition*	F004
Opioid system (mu/kappa)	Context-dependent	Human naltrexone data mixed — blocks benefit in one RCT, not in others	F005, F008
Lateral habenula NMDAR burst-firing	Disputed	Contested; replications fail	F011
Frontoparietal gamma / cortical E-I engagement (biomarker)	Readout observed	Correlative only — tracks response but not causal	F010

The canonical E/I mechanism is stated as the mechanistic basis for the approved drug:

> *"N-methyl-D-aspartate (NMDA) receptor antagonism on inhibitory interneurons disinhibits glutamate release and modulates downstream synaptic plasticity pathways, providing a mechanistic basis for the rapid antidepressant effects of esketamine."* — PMID: 41923438
>
> *"NMDA receptor antagonistic action causes glutamate-AMPA activation to spike, which in turn activates the BDNF-mTORC1 signalling pathway, promoting fast synaptogenesis"* — PMID: 42613710

Ketamine's action on its primary target (NMDAR antagonism) is not in doubt. What is in doubt is whether the *therapeutic benefit depends on* the downstream E/I shift. That is the core question, addressed in Section 3.

3. Evidence That a Mechanism *Mediates* the Benefit

This is the crux. The requirement is not that ketamine *also* shifts E/I balance, but that the antidepressant effect **depends on** it. The evidence here is strikingly mixed and, on balance, does not establish dependence.

3.1 Dissociation evidence: NMDAR-engaging analogues lack the benefit (refutes sufficiency)

If the E/I shift via NMDAR antagonism were sufficient, other drugs that block NMDARs should also be antidepressants. They are not.

Memantine (low-to-moderate affinity NMDA antagonist):

> *"the low-to-moderate-affinity NMDA antagonist memantine in doses of 5-20 mg/day was not effective in the treatment of major depressive disorder"* — PMID: 16390905

A separate augmentation RCT (β = 0.133 favoring placebo, p = 0.74) confirmed no benefit (PMID: 24229746).

Lanicemine (AZD6765), an NMDA channel blocker:

> *"neither dose was superior to placebo in reducing depressive symptoms on the primary end point or any secondary measures"* — PMID: 27681442 (phase IIb RCT, n = 302)

And in a head-to-head pharmacological-MRI study where lanicemine engaged NMDARs with a time course similar to ketamine:

> *"There was no significant antidepressant effect of either infusion compared to saline."* — PMID: 27133029

Interpretation: NMDAR engagement — and the E/I shift it should produce — is demonstrably *insufficient* for antidepressant benefit. This is a strong "analogue hits the target without the benefit" result.

3.2 The HNK metabolite: benefit without NMDAR inhibition (refutes necessity of NMDAR step)

Zanos et al. (2016, *Nature*) showed in mice that metabolism of ketamine to (2R,6R)-hydroxynorketamine is essential for antidepressant-like effects, and that these actions operate *outside* NMDAR inhibition:

> *"These antidepressant actions are independent of NMDAR inhibition but involve early and sustained activation of AMPARs"* — PMID: 27144355

Interpretation: This dissociates the antidepressant-related actions from the initiating NMDAR-antagonism step of the E/I model (animal model; the clinical relevance of HNK in humans remains debated, but the mechanistic point stands).

3.3 Direct in-patient E/I readouts are null (fails target-engagement test)

The most direct test of the E/I mechanism in patients is measuring GABA and glutamate with ^{1}H-MRS. A 2026 systematic review/meta-analysis of 11 such studies found no reliable effect:

> *"Meta-analyses from limited evidence base did not demonstrate statistically significant treatment-related alterations in any metabolite."* — PMID: 42768091

The same authors caution that the field's E/I index is not even a validated physiological quantity:

> *"the E=Glx/Glu and I = GABA framework adopted here, as used commonly in the field, is an assumed convention rather than a validated physiological equivalence"* — PMID: 42768091

Interpretation: In patients, ketamine does not reliably move the metabolites that define E/I imbalance, and the index itself is conceptually unvalidated.

3.4 Dissociation does not track response (the E/I signature is decoupled from benefit)

The acute dissociative state is the most direct behavioral readout of NMDAR antagonism / cortical E-I disruption. A 2024 review notes:

> *"The severity of treatment-emergent dissociative symptoms is typically not associated with a response"* — PMID: 39613748

Interpretation: If the therapeutic effect were driven by the NMDAR/E-I perturbation, the intensity of that perturbation (dissociation) should correlate with benefit. It does not.

3.5 Partial support: frontoparietal gamma tracks response

Not all E/I-related readouts are null. Gamma oscillations are an E/I-balance-dependent cortical signal:

> *"Pre-treatment versus post-treatment increases in gamma power in frontoparietal brain regions, observed in electroencephalogram (EEG) studies, is a promising brain-based biomarker of response to ketamine"* — PMID: 39613748

Interpretation: This is genuine partial support for cortical E/I *engagement* tracking response — but it is correlative, not a causal dissection.

3.6 Downstream BDNF pathway does not track response in patients

The E/I model's downstream effector is BDNF. In mice, loss-of-function evidence supports BDNF-dependence:

> *"ketamine and other NMDAR antagonists produce fast-acting behavioural antidepressant-like effects in mouse models, and that these effects depend on the rapid synthesis of brain-derived neurotrophic factor"* — PMID: 21677641
>
> *"This potentiation requires protein synthesis, brain-derived neurotrophic factor expression, eukaryotic elongation factor-2 kinase function, and increased surface expression of AMPA receptors."* — PMID: 23595756

But in patients, the pathway readout does not track clinical response. In an RCT of oral esketamine in TRD (n = 54):

> *"The change in BDNF levels during treatment did not significantly differ between the esketamine and placebo group during treatment or washout."* — PMID: 42462931
>
> *"In the esketamine arm, BDNF changes did not correlate with changes in depression severity"* — PMID: 42462931

Interpretation: The animal mechanism does not translate to a measurable patient-level pathway readout that tracks benefit.

3.7 Competing mechanism — opioid system (human dissection, mixed)

The most direct *human pharmacological dissection* used naltrexone to block opioid receptors. Williams et al. (2018, *Am J Psychiatry*, TRD crossover RCT):

> *"Because naltrexone dramatically blocked the antidepressant but not the dissociative effects of ketamine, the trial was halted"* — PMID: 30153752

A secondary analysis replicated opioid-dependence for antisuicidal effects:

> *"naltrexone attenuated the antisuicidality effects of ketamine on all three suicidality scales/subscales (linear mixed model, fixed pretreatment effect, $p < 0.01$)"* — PMID: 31467392

Critically, naltrexone blocked the *benefit* while *sparing dissociation* (the E/I signature) — the opposite of what the E/I model predicts. However, the opioid mechanism is itself unsettled: a 2026 systematic review concludes it is not primary:

> *"the opioid system likely functions as a context-dependent modulator rather than a primary mediator, particularly at standard antidepressant doses"* — PMID: 41607072

And a 2026 randomized crossover rCBF neuroimaging study in MDD found ketamine's central effects survived opioid blockade:

> *"the effects were not attenuated by naltrexone"* — PMID: 41922981

Interpretation: The opioid system is a genuine competing candidate with direct human dissection evidence, but it too is disputed. Its chief importance here is that it dissociates *therapeutic benefit* from the *E/I/dissociation signature*.

3.8 Competing mechanism — lateral habenula burst firing (contested)

A circuit-level alternative locates the critical action in the lateral habenula (LHb):

> *"its rapid antidepressant effects are critically dependent on ketamine's action in the lateral habenula"* — PMID: 30823984

But this is contested on two fronts. Replication failed for sustained effects:

> *"the sustained antidepressant-like effect of ketamine may not depend on burst firing of LHb neurons"* — PMID: 33664132

And the NMDAR-dependence of the burst firing itself was refuted:

> *"NMDARs are not necessary for the generation of burst firing in LHb neuro[ns]"* — PMID: 42368821

Interpretation: A competing circuit mechanism exists but is itself unresolved and partly refuted.

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4. Explicit Statements of Uncertainty

Multiple independent, recent reviews and protocols state plainly that ketamine's antidepressant mechanism is *not* settled:

> *"However, its specific mechanism of action remains unclear."* — 2025 ketamine biomarker review, PMID: 40780612

> *"While early studies show swift mood improvements, its precise mechanisms remain unclear."* — 2024 narrative review "Beyond NMDA Receptors," PMID: 39769420
>
> *"their mechanisms of action-the understanding of which is essential for refining therapeutic precision-remain unclear"* — 2026 adenosine study, PMID: 41193806
>
> *"While its clinical efficacy and safety are proven, the mechanisms underlying its antidepressant effect remain unclear."* — ReDREAM esketamine metabolomics protocol, PMID: 39360294

No source in the reviewed literature states that the E/I mechanism in MDD is *established*. The closest affirmative statements (e.g., PMID: 41923438) describe the disinhibition model as providing "a mechanistic basis" — i.e., a proposed/plausible mechanism, not a demonstrated causal one.

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5. Assessment

Classification: SEVERAL COMPETING CANDIDATES (with E/I imbalance a leading hypothesis that lacks causal confirmation in patients).

The clinical benefit is established. The mechanism is not. The E/I-imbalance account is the dominant hypothesis and has partial, indirect support (frontoparietal gamma tracking response; preclinical BDNF/AMPA dependence). But the body of refuting and dissociating evidence is substantial and directly addresses the "does the benefit *depend* on it" question:

- NMDAR-engaging analogues (memantine, lanicemine) fail → E/I shift is **insufficient**.
- The HNK metabolite reproduces antidepressant actions **independent of NMDAR inhibition** → the initiating NMDAR step is not strictly necessary.
- Patient MRS GABA/Glu/Glx readouts are **null**, and the E/I index is **unvalidated**.
- Dissociation (the acute E/I signature) **does not track response**.
- Naltrexone can block the benefit while **sparing dissociation** → benefit dissociable from the E/I signature.
- The downstream BDNF readout **does not track response in patients**.

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E/I-IMBALANCE MECHANISM — STATUS OF EACH LINK IN THE CHAIN
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NMDAR antagonism  ➝  disinhibition  ➝  glutamate/AMPA  ➝  BDNF-mTORC1  ➝  benefit
     |                    |                  |                |             |
[established]       [null in patient    [MRS null;        [not tracking   [ESTABLISHED
but analogues        MRS; dissociation   HNK bypasses      response in      clinically]
that engage it       decoupled from      NMDAR step]       patients]
FAIL clinically]     response]
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Conclusion: every intermediate link is either unconfirmed in patients
or actively dissociable from the therapeutic endpoint.
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Recommendation for the knowledge base: record "Excitation-Inhibition Imbalance" as **one of several candidate mechanisms / an open knowledge gap**, NOT as the known mechanism.

What experiment would most directly settle it?

A causal, patient-level dissection is needed — not another correlation. The most decisive designs would be:

1. **Pathway-blockade RCT in patients:** administer ketamine with and without a selective blocker of a proposed downstream node (e.g., an AMPA-receptor antagonist such as perampanel, or a selective interneuron/disinhibition manipulation such as an α5-GABA-A modulator) and test whether antidepressant response is abolished while confirming NMDAR target engagement. This is the human analogue of the mouse loss-of-function studies.
2. **Simultaneous multimodal target engagement vs. response:** within a single RCT, measure ^{1}H-MRS GABA/Glu/Glx, EEG gamma, and dissociation, and test *prospectively* which (if any) formally mediates the clinical response — rather than merely correlating.
3. **Head-to-head "mechanism-matched" comparators:** test an agent that produces the cortical E/I shift without ketamine's other actions (and vice versa) with adequate power, extending the memantine/lanicemine logic.

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6. Evidence Refuting "Excitation-Inhibition Imbalance" (given equal prominence)

Per the task's explicit instruction, the refuting evidence is consolidated here with the same care as the supporting evidence. Five independent lines argue against E/I imbalance *mediating* the benefit in MDD:

#	Refuting evidence	Model / population	Citation (PMID + quote)
R1	NMDA antagonist memantine is **not effective** in MDD	Human RCT (n = 32)	16390905: *"memantine in doses of 5-20 mg/day was not effective in the treatment of major depressive disorder"*
R2	NMDA channel blocker lanicemine **not superior to placebo** despite target engagement	Human RCT (n = 302); phMRI	27681442: *"neither dose was superior to placebo"*; 27133029: *"no significant antidepressant effect of either infusion compared to saline"*
R3	HNK metabolite antidepressant actions are **independent of NMDAR inhibition**	Mouse	27144355: *"independent of NMDAR inhibition but involve early and sustained activation of AMPARs"*
R4	Patient MRS shows **no reliable GABA/Glu/Glx change**; E/I index **unvalidated**	Human meta-analysis (11 studies)	42768091: *"did not demonstrate statistically significant treatment-related alterations in any metabolite"*; *"an assumed convention rather than a validated physiological equivalence"*
R5	Dissociation (E/I signature) **does not track response**; naltrexone blocks benefit but **spares dissociation**	Human	39613748: *"severity of treatment-emergent dissociative symptoms is typically not associated with a response"*; 30153752: *"naltrexone dramatically blocked the antidepressant but not the dissociative effects of ketamine"*

Supporting evidence for E/I engagement (for balance): frontoparietal gamma power increases are a promising response biomarker (PMID: 39613748), and preclinical loss-of-function studies show BDNF/AMPA-dependence downstream of NMDAR blockade (PMID: 21677641, PMID: 23595756). This support is correlative (gamma) or preclinical (BDNF), and the preclinical pathway does not translate to a patient-level readout that tracks response (PMID: 42462931).

Evidence Base (Key Literature)

PMID	Role	What it shows
42155204	Context	Oral-ketamine meta-analysis: SMD −0.62, p<0.00001 (efficacy established)
41923438	Mechanism statement	States disinhibition/E-I as the "mechanistic basis"; esketamine FDA-approved
42613710	Mechanism statement	Glutamate-AMPA-BDNF-mTORC1 cascade described
42768091	**Refutes**	Patient MRS meta-analysis null; E/I index unvalidated
27144355	**Refutes**	HNK antidepressant actions independent of NMDAR inhibition (mouse)
16390905	**Refutes**	Memantine not effective in MDD
27681442	**Refutes**	Lanicemine not superior to placebo (n=302)
27133029	**Refutes**	Lanicemine no antidepressant effect despite NMDAR engagement
30153752	Competing / dissociates	Naltrexone blocks benefit, spares dissociation
31467392	Competing	Naltrexone attenuates antisuicidal effect (p<0.01)
41607072	Competing (bounded)	Opioid system a modulator, not primary mediator
41922981	Bounds opioid claim	rCBF effects not attenuated by naltrexone
21677641	Supports (preclinical)	Antidepressant-like effect depends on BDNF synthesis (mouse)
23595756	Supports (preclinical)	Requires BDNF, eEF2K, surface AMPARs
42462931	**Refutes (patient)**	BDNF not changed vs placebo; not correlated with response
39613748	Mixed	

PMID	Role	What it shows
		Gamma tracks response (support); dissociation does not (refute)
30823984	Competing	LHb burst-firing mechanism proposed
33664132	Contests competitor	Sustained effect may not depend on LHb burst firing
42368821	Contests competitor	NMDARs not necessary for LHb burst firing
40780612, 39769420, 41193806, 39360294	Uncertainty	Four independent reviews/protocols: mechanism "remains unclear"

Limitations and Knowledge Gaps

1. **Preclinical-to-clinical gap.** The strongest mechanistic (loss-of-function) evidence — BDNF-dependence, HNK's NMDAR-independence — is from rodents at doses whose clinical relevance is debated. The best patient-level tests (MRS, BDNF) are correlational and null.
2. **No human causal dissection of the E/I step.** There is no patient study that blocks the proposed E/I/disinhibition pathway and shows the antidepressant response is abolished. The only human loss-of-function dissection (naltrexone) targets the *opioid* system, not E/I.
3. **Measurement validity.** The MRS-based E/I index (Glx/Glu for E, GABA for I) is, by the field's own admission, "an assumed convention rather than a validated physiological equivalence" (PMID: 42768091). Null MRS results could reflect insensitive measurement rather than absence of effect. "E/I imbalance" is also operationalized inconsistently across studies (MRS, EEG gamma, TMS cortical excitability, interneuron physiology), complicating adjudication.
4. **Small samples / heterogeneity.** The BDNF RCT (n=54), the naltrexone trial (n=12 completers), and MRS studies are individually small; dosing, timing, and enantiomer (racemic ketamine vs esketamine) vary across studies.
5. **Candidate mechanisms are not mutually exclusive.** Convergent serotonergic-glutamatergic, opioid, adenosine, and gut-metabolite pathways may each contribute; the question may not have a single answer.

6. **This review is literature-based.** No primary data were analyzed; conclusions rest on published abstracts and cited snippets, and some nuance resides in full texts not quoted here.

Proposed Follow-up Experiments / Actions

1. **Human pathway-blockade RCT (highest priority).** Ketamine ± a downstream-pathway blocker (e.g., AMPA antagonist, or α5-GABA-A modulator targeting the disinhibition step) in MDD/TRD, with confirmed NMDAR target engagement, testing whether response is abolished. This is the decisive causal test the field lacks.
2. **Prospective multimodal mediation study.** Within one RCT, measure MRS GABA/Glu/Glx, EEG frontoparietal gamma, and dissociation at matched timepoints; apply formal mediation analysis to clinical response. Determines which readout, if any, is on the causal path.
3. **Validate the E/I index.** Calibrate MRS-derived Glx/GABA against an independent physiological E/I measure (e.g., TMS-EEG cortical inhibition) before interpreting null results as absence of mechanism.
4. **Adequately powered comparator trials.** Extend the memantine/lanicemine logic: test agents that produce the cortical E/I shift without ketamine's full pharmacology, and HNK-like agents that bypass NMDAR, in head-to-head MDD trials.
5. **Knowledge-base action.** Reclassify the ketamine→"Excitation-Inhibition Imbalance" link in MDD from *known mechanism* to **candidate mechanism** / **open knowledge gap**, cross-linking the competing candidates (BDNF/AMPA synaptogenesis, NMDAR-independent HNK, opioid modulation, LHb circuit, serotonergic/adenosine mechanisms) and the explicit-uncertainty citations.

Report compiled from 12 confirmed findings and 49 reviewed papers over 5 investigation iterations. Every claim above is tied to a primary-literature PMID and a verbatim abstract quote; preclinical versus human evidence is distinguished throughout.
